



Built-In Functions and Operators

Table of Contents

-  Operators
-  String Functions
-  Numeric Functions
-  Date and Time Functions
-  Aggregate Functions
-  Window Functions
-  JSON Functions
-  Spatial Functions
-  Comparison Functions
-  Control Flow Functions
-  Information Functions
-  Encryption and Compression
-  Locking Functions
-  Regular Expression Functions
-  Replication Functions
-  XML Functions
-  Miscellaneous Functions

Operators

Name	Description	Usage Example
&	Bitwise AND	<code>SELECT a & b;</code>
>	Greater than operator	<code>SELECT a > b;</code>
>>	Right shift	<code>SELECT a >> b;</code>
>=	Greater than or equal operator	<code>SELECT a >= b;</code>
<	Less than operator	<code>SELECT a < b;</code>
<>, !=	Not equal operator	<code>SELECT a <> b;</code>
<<	Left shift	<code>SELECT a << b;</code>
<=	Less than or equal operator	<code>SELECT a <= b;</code>
<=>	NULL-safe equal to operator	<code>SELECT a <=> b;</code>
%, MOD	Modulo operator	<code>SELECT a % b;</code>
*	Multiplication operator	<code>SELECT a * b;</code>
+	Addition operator	<code>SELECT a + b;</code>

Name	Description	Usage Example
-	Minus operator	<code>SELECT a - b;</code>
- (unary)	Change the sign of the argument	<code>SELECT -a;</code>
->	Return value from JSON column after evaluating path; equivalent to JSON_EXTRACT().	<code>SELECT a -> '\$.x';</code>
->>	Return value from JSON column after evaluating path and unquoting the result; equivalent to JSON_UNQUOTE(JSON_EXTRACT()).	<code>SELECT a ->> '\$.x';</code>
/	Division operator	<code>SELECT a / b;</code>
:=	Assign a value	<code>SET @a := 1;</code>
=	Assign a value (as part of a SET statement, or as part of the SET clause in an UPDATE statement)	<code>UPDATE my_table SET col = 1 WHERE id = 1;</code>
=	Equal operator	<code>SELECT a = b;</code>
^	Bitwise XOR	<code>SELECT a ^ b;</code>
	Bitwise OR	<code>SELECT a \ b;</code>
~	Bitwise inversion	<code>SELECT ~a;</code>
AND, &&	Logical AND	<code>SELECT a AND b;</code>
OR,	Logical OR	<code>SELECT a OR b;</code>
NOT, !	Negates value	<code>SELECT NOT a;</code>
XOR	Logical XOR	<code>SELECT a XOR b;</code>
BETWEEN ... AND ...	Whether a value is within a range of values	<code>SELECT * FROM t WHERE x BETWEEN 1 AND 10;</code>
NOT BETWEEN ... AND ...	Whether a value is not within a range of values	<code>SELECT * FROM t WHERE x NOT BETWEEN 1 AND 10;</code>
IN()	Whether a value is within a set of values	<code>SELECT * FROM t WHERE x IN (1,2,3);</code>
NOT IN()	Whether a value is not within a set of values	<code>SELECT * FROM t WHERE x NOT IN (1,2,3);</code>
IS	Test a value against a boolean	<code>SELECT a IS TRUE;</code>
IS NOT	Test a value against a boolean	<code>SELECT a IS NOT TRUE;</code>
IS NULL	NULL value test	<code>SELECT * FROM t WHERE col IS NULL;</code>
IS NOT NULL	NOT NULL value test	<code>SELECT * FROM t WHERE col IS NOT NULL;</code>
LIKE	Simple pattern matching	<code>SELECT * FROM t WHERE col LIKE 'abc%';</code>
NOT LIKE	Negation of simple pattern matching	<code>SELECT * FROM t WHERE col NOT LIKE 'abc%';</code>
REGEXP	Whether string matches regular expression	<code>SELECT col REGEXP '^[0-9]+\$';</code>
NOT REGEXP	Negation of REGEXP	<code>SELECT col NOT REGEXP '^[0-9]+\$';</code>
RLIKE	Whether string matches regular expression	<code>SELECT col RLIKE '^[0-9]+\$';</code>
DIV	Integer division	<code>SELECT 7 DIV 2;</code>
CASE	Case operator	<code>SELECT CASE WHEN a > 0 THEN 'yes' ELSE 'no' END;</code>
INTERVAL()	Return the index of the argument that is less than the first argument	<code>SELECT INTERVAL(10, 1, 5, 15);</code>

Name	Description	Usage Example
MEMBER OF()	Returns true (1) if first operand matches any element of JSON array passed as second operand, otherwise returns false (0)	<pre>SELECT 'a' MEMBER OF (JSON_ARRAY('a','b'));</pre>

abc

String Functions

Name	Description	Usage Example
ASCII()	Return numeric value of left-most character	<pre>SELECT ASCII('A');</pre>
BIN()	Return a string containing binary representation of a number	<pre>SELECT BIN(7);</pre>
BIT_LENGTH()	Return length of argument in bits	<pre>SELECT BIT_LENGTH('abc');</pre>
CHAR()	Return the character for each integer passed	<pre>SELECT CHAR(65);</pre>
CHAR_LENGTH()	Return number of characters in argument	<pre>SELECT CHAR_LENGTH('abc');</pre>
CHARACTER_LENGTH()	Synonym for CHAR_LENGTH()	<pre>SELECT CHARACTER_LENGTH('abc');</pre>
CHARSET()	Return the character set of the argument	<pre>SELECT CHARSET('abc');</pre>
COERCIBILITY()	Return the collation coercibility value of the string argument	<pre>SELECT COERCIBILITY('abc');</pre>
COLLATION()	Return the collation of the string argument	<pre>SELECT COLLATION('abc');</pre>
CONCAT()	Return concatenated string	<pre>SELECT CONCAT('a','b');</pre>
CONCAT_WS()	Return concatenate with separator	<pre>SELECT CONCAT_WS('-', 'a', 'b');</pre>
ELT()	Return string at index number	<pre>SELECT ELT(2, 'a','b','c');</pre>
EXPORT_SET()	Return a string such that for every bit set in the value bits, you get an on string and for every unset bit, you get an off string	<pre>SELECT EXPORT_SET(5,'Y','N');</pre>
FIELD()	Index (position) of first argument in subsequent arguments	<pre>SELECT FIELD('b','a','b','c');</pre>
FIND_IN_SET()	Index (position) of first argument within second argument	<pre>SELECT FIND_IN_SET('b','a,b,c');</pre>
FORMAT()	Return a number formatted to specified number of decimal places	<pre>SELECT FORMAT(1234.56,2);</pre>
HEX()	Hexadecimal representation of decimal or string value	<pre>SELECT HEX('abc');</pre>
INSERT()	Insert substring at specified position up to specified number of characters	<pre>SELECT INSERT('abc',2,1,'Z');</pre>
INSTR()	Return the index of the first occurrence of substring	<pre>SELECT INSTR('abc','b');</pre>
LCASE()	Synonym for LOWER()	<pre>SELECT LCASE('ABC');</pre>
LEFT()	Return the leftmost number of characters as specified	<pre>SELECT LEFT('abc',2);</pre>
LENGTH()	Return the length of a string in bytes	<pre>SELECT LENGTH('abc');</pre>
LOAD_FILE()	Load the named file	<pre>SELECT LOAD_FILE('/path/file');</pre>
LOCATE()	Return the position of the first occurrence of substring	<pre>SELECT LOCATE('b','abc');</pre>
LOWER()	Return the argument in lowercase	<pre>SELECT LOWER('ABC');</pre>
LPAD()	Return the string argument, left-padded with the specified string	<pre>SELECT LPAD('abc',5,'0');</pre>
LTRIM()	Remove leading spaces	<pre>SELECT LTRIM(' abc');</pre>
MAKE_SET()	Return a set of comma-separated strings that have the corresponding bit in bits set	<pre>SELECT MAKE_SET(5,'a','b','c');</pre>
MID()	Return a substring starting from the specified position	<pre>SELECT MID('abc',2,1);</pre>
OCT()	Return a string containing octal representation of a number	<pre>SELECT OCT(8);</pre>
OCTET_LENGTH()	Synonym for LENGTH()	<pre>SELECT OCTET_LENGTH('abc');</pre>

Name	Description	Usage Example
ORD()	Return character code for leftmost character of the argument	<code>SELECT ORD('A');</code>
POSITION()	Synonym for LOCATE()	<code>SELECT POSITION('b' IN 'abc');</code>
QUOTE()	Escape the argument for use in an SQL statement	<code>SELECT QUOTE("I'm");</code>
REPEAT()	Repeat a string the specified number of times	<code>SELECT REPEAT('a',3);</code>
REPLACE()	Replace occurrences of a specified string	<code>SELECT REPLACE('abc','b','Z');</code>
REVERSE()	Reverse the characters in a string	<code>SELECT REVERSE('abc');</code>
RIGHT()	Return the specified rightmost number of characters	<code>SELECT RIGHT('abc',2);</code>
RPAD()	Append string the specified number of times	<code>SELECT RPAD('abc',5,'0');</code>
RTRIM()	Remove trailing spaces	<code>SELECT RTRIM('abc ');</code>
SOUNDEX()	Return a soundex string	<code>SELECT SOUNDEX('abc');</code>
SOUNDS LIKE	Compare sounds	<code>SELECT 'Robert' SOUNDS LIKE 'Rupert';</code>
SPACE()	Return a string of the specified number of spaces	<code>SELECT SPACE(3);</code>
STRCMP()	Compare two strings	<code>SELECT STRCMP('a','b');</code>
SUBSTR()	Return the substring as specified	<code>SELECT SUBSTR('abc',2);</code>
SUBSTRING()	Return the substring as specified	<code>SELECT SUBSTRING('abc',2);</code>
SUBSTRING_INDEX()	Return a substring from a string before the specified number of occurrences of the delimiter	<code>SELECT SUBSTRING_INDEX('a,b,c',' ',2);</code>
TRIM()	Remove leading and trailing spaces	<code>SELECT TRIM(' abc ');</code>
UCASE()	Synonym for UPPER()	<code>SELECT UCASE('abc');</code>
UNHEX()	Return a string containing hex representation of a number	<code>SELECT UNHEX('4D');</code>
UPPER()	Convert to uppercase	<code>SELECT UPPER('abc');</code>
WEIGHT_STRING()	Return the weight string for a string	<code>SELECT WEIGHT_STRING('abc');</code>

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Numeric Functions

Name	Description	Usage Example
ABS()	Return the absolute value	<code>SELECT ABS(-5);</code>
ACOS()	Return the arc cosine	<code>SELECT ACOS(0.5);</code>
ASIN()	Return the arc sine	<code>SELECT ASIN(0.5);</code>
ATAN()	Return the arc tangent	<code>SELECT ATAN(1);</code>
ATAN2()	Return the arc tangent of the two arguments	<code>SELECT ATAN2(1,1);</code>
CEIL()	Return the smallest integer value not less than the arg	<code>SELECT CEIL(1.2);</code>
CEILING()	Return the smallest integer value not less than the arg	<code>SELECT CEILING(1.2);</code>
CONV()	Convert numbers between different number bases	<code>SELECT CONV('a',16,10);</code>
COS()	Return the cosine	<code>SELECT COS(0);</code>
COT()	Return the cotangent	<code>SELECT COT(1);</code>
CRC32()	Compute a cyclic redundancy check value	<code>SELECT CRC32('abc');</code>
DEGREES()	Convert radians to degrees	<code>SELECT DEGREES(PI());</code>

Name	Description	Usage Example
DISTANCE()	Calculates the distance between two vectors	<code>SELECT DISTANCE(v1, v2);</code>
EXP()	Raise to the power of	<code>SELECT EXP(1);</code>
FLOOR()	Return the largest integer value not greater than the arg	<code>SELECT FLOOR(1.9);</code>
LN()	Return the natural logarithm of the argument	<code>SELECT LN(2);</code>
LOG()	Return the natural logarithm of the first argument	<code>SELECT LOG(2);</code>
LOG10()	Return the base-10 logarithm of the argument	<code>SELECT LOG10(100);</code>
LOG2()	Return the base-2 logarithm of the argument	<code>SELECT LOG2(8);</code>
MOD()	Return the remainder	<code>SELECT MOD(7,2);</code>
PI()	Return the value of pi	<code>SELECT PI();</code>
POW()	Return the argument raised to the specified power	<code>SELECT POW(2,3);</code>
POWER()	Return the argument raised to the specified power	<code>SELECT POWER(2,3);</code>
RADIANS()	Return argument converted to radians	<code>SELECT RADIANS(180);</code>
RAND()	Return a random floating-point value	<code>SELECT RAND();</code>
ROUND()	Round the argument	<code>SELECT ROUND(1.234,2);</code>
SIGN()	Return the sign of the argument	<code>SELECT SIGN(-5);</code>
SIN()	Return the sine of the argument	<code>SELECT SIN(0);</code>
SQRT()	Return the square root of the argument	<code>SELECT SQRT(9);</code>
TAN()	Return the tangent of the argument	<code>SELECT TAN(1);</code>
TRUNCATE()	Truncate to specified number of decimal places	<code>SELECT TRUNCATE(1.234,2);</code>



Date and Time Functions

Name	Description	Usage Example
ADDDATE()	Add time values (intervals) to a date value	<code>SELECT ADDDATE('2024-01-01', INTERVAL 1 DAY);</code>
ADDTIME()	Add time	<code>SELECT ADDTIME('10:00:00', '01:00:00');</code>
CURDATE()	Return the current date	<code>SELECT CURDATE();</code>
CURRENT_DATE(), CURRENT_DATE	Synonyms for CURDATE()	<code>SELECT CURRENT_DATE();</code>
CURRENT_TIME(), CURRENT_TIME	Synonyms for CURTIME()	<code>SELECT CURRENT_TIME();</code>
CURRENT_TIMESTAMP(), CURRENT_TIMESTAMP	Synonyms for NOW()	<code>SELECT CURRENT_TIMESTAMP();</code>
CURTIME()	Return the current time	<code>SELECT CURTIME();</code>
DATE()	Extract the date part of a date or datetime expression	<code>SELECT DATE(NOW());</code>
DATE_ADD()	Add time values (intervals) to a date value	<code>SELECT DATE_ADD('2024-01-01', INTERVAL 7 DAY);</code>
DATE_FORMAT()	Format date as specified	<code>SELECT DATE_FORMAT(NOW(), '%Y-%m-%d');</code>
DATE_SUB()	Subtract a time value (interval) from a date	<code>SELECT DATE_SUB('2024-01-01', INTERVAL 7 DAY);</code>

Name	Description	Usage Example
DATEDIFF()	Subtract two dates	<code>SELECT DATEDIFF('2024-01-10','2024-01-01');</code>
DAY()	Synonym for DAYOFMONTH()	<code>SELECT DAY('2024-01-05');</code>
DAYNAME()	Return the name of the weekday	<code>SELECT DAYNAME('2024-01-05');</code>
DAYOFMONTH()	Return the day of the month (0-31)	<code>SELECT DAYOFMONTH('2024-01-05');</code>
DAYOFWEEK()	Return the weekday index of the argument	<code>SELECT DAYOFWEEK('2024-01-05');</code>
DAYOFYEAR()	Return the day of the year (1-366)	<code>SELECT DAYOFYEAR('2024-01-05');</code>
EXTRACT()	Extract part of a date	<code>SELECT EXTRACT(YEAR FROM NOW());</code>
FROM_DAYS()	Convert a day number to a date	<code>SELECT FROM_DAYS(738885);</code>
FROM_UNIXTIME()	Format Unix timestamp as a date	<code>SELECT FROM_UNIXTIME(1700000000);</code>
GET_FORMAT()	Return a date format string	<code>SELECT GET_FORMAT(DATE,'ISO');</code>
HOUR()	Extract the hour	<code>SELECT HOUR('10:11:12');</code>
LAST_DAY()	Return the last day of the month for the argument	<code>SELECT LAST_DAY('2024-01-05');</code>
LOCALTIME(), LOCALTIME	Synonym for NOW()	<code>SELECT LOCALTIME();</code>
LOCALTIMESTAMP, LOCALTIMESTAMP()	Synonym for NOW()	<code>SELECT LOCALTIMESTAMP();</code>
MAKEDATE()	Create a date from the year and day of year	<code>SELECT MAKEDATE(2024, 100);</code>
MAKETIME()	Create time from hour, minute, second	<code>SELECT MAKETIME(10,11,12);</code>
MICROSECOND()	Return the microseconds from argument	<code>SELECT MICROSECOND('10:11:12.123456');</code>
MINUTE()	Return the minute from the argument	<code>SELECT MINUTE('10:11:12');</code>
MONTH()	Return the month from the date passed	<code>SELECT MONTH('2024-01-05');</code>
MONTHNAME()	Return the name of the month	<code>SELECT MONTHNAME('2024-01-05');</code>
NOW()	Return the current date and time	<code>SELECT NOW();</code>
PERIOD_ADD()	Add a period to a year-month	<code>SELECT PERIOD_ADD(202401, 2);</code>
PERIOD_DIFF()	Return the number of months between periods	<code>SELECT PERIOD_DIFF(202403, 202401);</code>
QUARTER()	Return the quarter from a date argument	<code>SELECT QUARTER('2024-01-05');</code>
SEC_TO_TIME()	Converts seconds to 'hh:mm:ss' format	<code>SELECT SEC_TO_TIME(3661);</code>
SECOND()	Return the second (0-59)	<code>SELECT SECOND('10:11:12');</code>
STR_TO_DATE()	Convert a string to a date	<code>SELECT STR_TO_DATE('2024-01-05','%Y-%m-%d');</code>
SUBDATE()	Synonym for DATE_SUB() when invoked with three arguments	<code>SELECT SUBDATE('2024-01-05', INTERVAL 1 DAY);</code>
SUBTIME()	Subtract times	<code>SELECT SUBTIME('10:11:12','00:10:00');</code>
SYSDATE()	Return the time at which the function executes	<code>SELECT SYSDATE();</code>
TIME()	Extract the time portion of the expression passed	<code>SELECT TIME(NOW());</code>
TIME_FORMAT()	Format as time	<code>SELECT TIME_FORMAT(NOW(), '%H:%i:%s');</code>
TIME_TO_SEC()	Return the argument converted to seconds	<code>SELECT TIME_TO_SEC('01:01:01');</code>
TIMEDIFF()	Subtract time	<code>SELECT TIMEDIFF('10:11:12','10:10:10');</code>
TIMESTAMP()	With a single argument, returns date/datetime; with two, the sum of the arguments	<code>SELECT TIMESTAMP('2024-01-01','10:00:00');</code>
TIMESTAMPADD()	Add an interval to a datetime expression	<code>SELECT TIMESTAMPADD(DAY, 1, NOW());</code>

Name	Description	Usage Example
TIMESTAMPDIFF()	Return the difference of two datetime expressions, using the units specified	<code>SELECT TIMESTAMPDIFF(DAY, '2024-01-01', '2024-01-05');</code>
TO_DAYS()	Return the date argument converted to days	<code>SELECT TO_DAYS('2024-01-05');</code>
TO_SECONDS()	Return the date or datetime argument converted to seconds since Year 0	<code>SELECT TO_SECONDS('2024-01-05');</code>
UNIX_TIMESTAMP()	Return a Unix timestamp	<code>SELECT UNIX_TIMESTAMP();</code>
UTC_DATE()	Return the current UTC date	<code>SELECT UTC_DATE();</code>
UTC_TIME()	Return the current UTC time	<code>SELECT UTC_TIME();</code>
UTC_TIMESTAMP()	Return the current UTC date and time	<code>SELECT UTC_TIMESTAMP();</code>
WEEK()	Return the week number	<code>SELECT WEEK('2024-01-05');</code>
WEEKDAY()	Return the weekday index	<code>SELECT WEEKDAY('2024-01-05');</code>
WEEKOFYEAR()	Return the calendar week of the date (1-53)	<code>SELECT WEEKOFYEAR('2024-01-05');</code>
YEAR()	Return the year	<code>SELECT YEAR('2024-01-05');</code>
YEARWEEK()	Return the year and week	<code>SELECT YEARWEEK('2024-01-05');</code>

Aggregate Functions

Name	Description	Usage Example
AVG()	Return the average value of the argument	<code>SELECT AVG(col) FROM my_table;</code>
BIT_AND()	Return bitwise AND	<code>SELECT BIT_AND(col) FROM my_table;</code>
BIT_OR()	Return bitwise OR	<code>SELECT BIT_OR(col) FROM my_table;</code>
BIT_XOR()	Return bitwise XOR	<code>SELECT BIT_XOR(col) FROM my_table;</code>
COUNT()	Return a count of the number of rows returned	<code>SELECT COUNT(*) FROM my_table;</code>
COUNT(DISTINCT)	Return the count of a number of different values	<code>SELECT COUNT(DISTINCT col) FROM t;</code>
GROUP_CONCAT()	Return a concatenated string	<code>SELECT GROUP_CONCAT(col) FROM my_table;</code>
MAX()	Return the maximum value	<code>SELECT MAX(col) FROM my_table;</code>
MIN()	Return the minimum value	<code>SELECT MIN(col) FROM my_table;</code>
STD()	Return the population standard deviation	<code>SELECT STD(col) FROM my_table;</code>
STDDEV()	Return the population standard deviation	<code>SELECT STDDEV(col) FROM my_table;</code>
STDDEV_POP()	Return the population standard deviation	<code>SELECT STDDEV_POP(col) FROM my_table;</code>
STDDEV_SAMP()	Return the sample standard deviation	<code>SELECT STDDEV_SAMP(col) FROM my_table;</code>
SUM()	Return the sum	<code>SELECT SUM(col) FROM my_table;</code>
VAR_POP()	Return the population standard variance	<code>SELECT VAR_POP(col) FROM my_table;</code>
VAR_SAMP()	Return the sample variance	<code>SELECT VAR_SAMP(col) FROM my_table;</code>
VARIANCE()	Return the population standard variance	<code>SELECT VARIANCE(col) FROM my_table;</code>

Window Functions

Name	Description	Usage Example
CUME_DIST()	Cumulative distribution value	<code>SELECT CUME_DIST() OVER ();</code>
DENSE_RANK()	Rank of current row within its partition, without gaps	<code>SELECT DENSE_RANK() OVER ();</code>
FIRST_VALUE()	Value of argument from first row of window frame	<code>SELECT FIRST_VALUE(col) OVER ();</code>
GROUPING()	Distinguish super-aggregate ROLLUP rows from regular rows	<code>SELECT GROUPING(col) FROM t GROUP BY col WITH ROLLUP;</code>
LAG()	Value of argument from row lagging current row within partition	<code>SELECT LAG(col) OVER ();</code>
LAST_VALUE()	Value of argument from last row of window frame	<code>SELECT LAST_VALUE(col) OVER ();</code>
LEAD()	Value of argument from row leading current row within partition	<code>SELECT LEAD(col) OVER ();</code>
NTH_VALUE()	Value of argument from N-th row of window frame	<code>SELECT NTH_VALUE(col,2) OVER ();</code>
NTILE()	Bucket number of current row within its partition.	<code>SELECT NTILE(4) OVER ();</code>
PERCENT_RANK()	Percentage rank value	<code>SELECT PERCENT_RANK() OVER ();</code>
RANK()	Rank of current row within its partition, with gaps	<code>SELECT RANK() OVER ();</code>
ROW_NUMBER()	Number of current row within its partition	<code>SELECT ROW_NUMBER() OVER ();</code>

JSON Functions

Name	Description	Usage Example
JSON_ARRAY()	Create JSON array	<code>SELECT JSON_ARRAY(1,2,3);</code>
JSON_ARRAY_APPEND()	Append data to JSON document	<code>SELECT JSON_ARRAY_APPEND('[1]', '\$', 2);</code>
JSON_ARRAY_INSERT()	Insert into JSON array	<code>SELECT JSON_ARRAY_INSERT('[1]', '\$[1]', 2);</code>
JSON_ARRAYAGG()	Return result set as a single JSON array	<code>SELECT JSON_ARRAYAGG(col) FROM t;</code>
JSON_CONTAINS()	Whether JSON document contains specific object at path	<code>SELECT JSON_CONTAINS('{ "a":1}', '1', '\$.a');</code>
JSON_CONTAINS_PATH()	Whether JSON document contains any data at path	<code>SELECT JSON_CONTAINS_PATH('{ "a":1}', 'one', '\$.a');</code>
JSON_DEPTH()	Maximum depth of JSON document	<code>SELECT JSON_DEPTH('{ "a":1}');</code>
JSON_DUALITY_OBJECT()	Create JSON duality object	<code>SELECT JSON_DUALITY_OBJECT();</code>
JSON_EXTRACT()	Return data from JSON document	<code>SELECT JSON_EXTRACT('{ "a":1}', '\$.a');</code>
JSON_INSERT()	Insert data into JSON document	<code>SELECT JSON_INSERT('{ "a":1}', '\$.b', 2);</code>
JSON_KEYS()	Array of keys from JSON document	<code>SELECT JSON_KEYS('{ "a":1, "b":2}');</code>
JSON_LENGTH()	Number of elements in JSON document	<code>SELECT JSON_LENGTH('[1,2,3]');</code>
JSON_MERGE()	Merge JSON documents, preserving duplicate keys.	<code>SELECT JSON_MERGE('{ "a":1}', '{ "a":2}');</code>

Name	Description	Usage Example
	Deprecated synonym for JSON_MERGE_PRESERVE()	
JSON_MERGE_PATCH()	Merge JSON documents, replacing values of duplicate keys	<pre>SELECT JSON_MERGE_PATCH('{ "a":1}', '{ "a":2}');</pre>
JSON_MERGE_PRESERVE()	Merge JSON documents, preserving duplicate keys	<pre>SELECT JSON_MERGE_PRESERVE('{ "a":1}', '{ "a":2}');</pre>
JSON_OBJECT()	Create JSON object	<pre>SELECT JSON_OBJECT('a',1);</pre>
JSON_OBJECTAGG()	Return result set as a single JSON object	<pre>SELECT JSON_OBJECTAGG(k,v) FROM t;</pre>
JSON_OVERLAPS()	Compare JSON documents; return TRUE if they overlap	<pre>SELECT JSON_OVERLAPS('{ "a":1}', '{ "a":2}');</pre>
JSON_PRETTY()	Print a JSON document in human-readable format	<pre>SELECT JSON_PRETTY('{ "a":1}');</pre>
JSON_QUOTE()	Quote JSON document	<pre>SELECT JSON_QUOTE('abc');</pre>
JSON_REMOVE()	Remove data from JSON document	<pre>SELECT JSON_REMOVE('{ "a":1}', '\$.a');</pre>
JSON_REPLACE()	Replace values in JSON document	<pre>SELECT JSON_REPLACE('{ "a":1}', '\$.a',2);</pre>
JSON_SCHEMA_VALID()	Validate JSON document against JSON schema	<pre>SELECT JSON_SCHEMA_VALID('{ "type":"object"}', '{ "a":1}');</pre>
JSON_SCHEMA_VALIDATION_REPORT()	Validate JSON document against JSON schema; returns report	<pre>SELECT JSON_SCHEMA_VALIDATION_REPORT('{ "type":"object"}', '{ "a":1}');</pre>
JSON_SEARCH()	Path to value within JSON document	<pre>SELECT JSON_SEARCH('{ "a":1}', 'one',1);</pre>
JSON_SET()	Insert data into JSON document	<pre>SELECT JSON_SET('{ "a":1}', '\$.b',2);</pre>
JSON_STORAGE_FREE()	Freed space within binary representation of JSON column value following partial update	<pre>SELECT JSON_STORAGE_FREE('{}');</pre>
JSON_STORAGE_SIZE()	Space used for storage of binary representation of a JSON document	<pre>SELECT JSON_STORAGE_SIZE('{}');</pre>
JSON_TABLE()	Return data from a JSON expression as a relational table	<pre>SELECT * FROM JSON_TABLE('[1,2]', '\$[*]' COLUMNS (x INT PATH '\$'));</pre>
JSON_TYPE()	Type of JSON value	<pre>SELECT JSON_TYPE('{ "a":1}');</pre>
JSON_UNQUOTE()	Unquote JSON value	<pre>SELECT JSON_UNQUOTE('"a"');</pre>
JSON_VALID()	Whether JSON value is valid	<pre>SELECT JSON_VALID('{ "a":1}');</pre>
JSON_VALUE()	Extract value from JSON document at path; return as VARCHAR(512) or specified type	<pre>SELECT JSON_VALUE('{ "a":1}', '\$.a');</pre>

Spatial Functions

Name	Description	Usage Example
GeomCollection()	Construct geometry collection from geometries	<code>SELECT GeomCollection();</code>
GeometryCollection()	Construct geometry collection from geometries	<code>SELECT GeometryCollection();</code>
LineString()	Construct LineString from Point values	<code>SELECT LineString();</code>
MultiLineString()	Construct MultiLineString from LineString values	<code>SELECT MultiLineString();</code>
MultiPoint()	Construct MultiPoint from Point values	<code>SELECT MultiPoint();</code>
MultiPolygon()	Construct MultiPolygon from Polygon values	<code>SELECT MultiPolygon();</code>
Point()	Construct Point from coordinates	<code>SELECT Point(1,1);</code>
Polygon()	Construct Polygon from LineString arguments	<code>SELECT Polygon();</code>
MBRContains()	Whether MBR of one geometry contains MBR of another	<code>SELECT MBRContains(a,b);</code>
MBRCoveredBy()	Whether one MBR is covered by another	<code>SELECT MBRCoveredBy(a,b);</code>
MBRCovers()	Whether one MBR covers another	<code>SELECT MBRCovers(a,b);</code>
MBRDisjoint()	Whether MBRs of two geometries are disjoint	<code>SELECT MBRDisjoint(a,b);</code>
MBREquals()	Whether MBRs of two geometries are equal	<code>SELECT MBREquals(a,b);</code>
MBRIntersects()	Whether MBRs of two geometries intersect	<code>SELECT MBRIntersects(a,b);</code>
MBROverlaps()	Whether MBRs of two geometries overlap	<code>SELECT MBROverlaps(a,b);</code>
MBRTouches()	Whether MBRs of two geometries touch	<code>SELECT MBRTouches(a,b);</code>
MBRWithin()	Whether MBR of one geometry is within MBR of another	<code>SELECT MBRWithin(a,b);</code>
ST_Area()	Return Polygon or MultiPolygon area	<code>SELECT ST_Area(g);</code>
ST_AsBinary(), ST_AsWKB()	Convert from internal geometry format to WKB	<code>SELECT ST_AsBinary(g);</code>
ST_AsGeoJSON()	Generate GeoJSON object from geometry	<code>SELECT ST_AsGeoJSON(g);</code>
ST_AsText(), ST_AsWKT()	Convert from internal geometry format to WKT	<code>SELECT ST_AsText(g);</code>

Name	Description	Usage Example
ST_Buffer()	Return geometry of points within given distance from geometry	<code>SELECT ST_Buffer(g, 10);</code>
ST_Buffer_Strategy()	Produce strategy option for ST_Buffer()	<code>SELECT ST_Buffer_Strategy('quad_segs=8');</code>
ST_Centroid()	Return centroid as a point	<code>SELECT ST_Centroid(g);</code>
ST_Collect()	Aggregate spatial values into collection	<code>SELECT ST_Collect(g);</code>
ST_Contains()	Whether one geometry contains another	<code>SELECT ST_Contains(a,b);</code>
ST_ConvexHull()	Return convex hull of geometry	<code>SELECT ST_ConvexHull(g);</code>
ST_Crosses()	Whether one geometry crosses another	<code>SELECT ST_Crosses(a,b);</code>
ST_Difference()	Return point set difference of two geometries	<code>SELECT ST_Difference(a,b);</code>
ST_Dimension()	Dimension of geometry	<code>SELECT ST_Dimension(g);</code>
ST_Disjoint()	Whether one geometry is disjoint from another	<code>SELECT ST_Disjoint(a,b);</code>
ST_Distance()	The distance of one geometry from another	<code>SELECT ST_Distance(a,b);</code>
ST_Distance_Sphere()	Minimum distance on earth between two geometries	<code>SELECT ST_Distance_Sphere(a,b);</code>
ST_EndPoint()	End Point of LineString	<code>SELECT ST_EndPoint(g);</code>
ST_Envelope()	Return MBR of geometry	<code>SELECT ST_Envelope(g);</code>
ST_Equals()	Whether one geometry is equal to another	<code>SELECT ST_Equals(a,b);</code>
ST_ExteriorRing()	Return exterior ring of Polygon	<code>SELECT ST_ExteriorRing(g);</code>
ST_FrechetDistance()	The discrete Fréchet distance of one geometry from another	<code>SELECT ST_FrechetDistance(a,b);</code>
ST_GeoHash()	Produce a geohash value	<code>SELECT ST_GeoHash(g);</code>
ST_GeomCollFromText(), ST_GeometryCollectionFromText(), ST_GeomCollFromTxt()	Return geometry collection from WKT	<code>SELECT ST_GeomCollFromText('GEOMETRYCOLLECTION()');</code>
ST_GeomCollFromWKB(), ST_GeometryCollectionFromWKB()	Return geometry collection from WKB	<code>SELECT ST_GeomCollFromWKB(g);</code>
ST_GeometryN()	Return N-th geometry from geometry collection	<code>SELECT ST_GeometryN(g,1);</code>
ST_GeometryType()	Return name of geometry type	<code>SELECT ST_GeometryType(g);</code>
ST_GeomFromGeoJSON()	Generate geometry from GeoJSON object	<code>SELECT ST_GeomFromGeoJSON('{ "type": "Point", "coordinates": [0,0] }');</code>
ST_GeomFromText(), ST_GeometryFromText()	Return geometry from WKT	<code>SELECT ST_GeomFromText('POINT(0 0)');</code>

Name	Description	Usage Example
ST_GeomFromWKB(), ST_GeometryFromWKB()	Return geometry from WKB	<code>SELECT ST_GeomFromWKB(g);</code>
ST_HausdorffDistance()	The discrete Hausdorff distance of one geometry from another	<code>SELECT ST_HausdorffDistance(a,b);</code>
ST_InteriorRingN()	Return N-th interior ring of Polygon	<code>SELECT ST_InteriorRingN(g,1);</code>
ST_Intersection()	Return point set intersection of two geometries	<code>SELECT ST_Intersection(a,b);</code>
ST_Intersects()	Whether one geometry intersects another	<code>SELECT ST_Intersects(a,b);</code>
ST_IsClosed()	Whether a geometry is closed and simple	<code>SELECT ST_IsClosed(g);</code>
ST_IsEmpty()	Whether a geometry is empty	<code>SELECT ST_IsEmpty(g);</code>
ST_IsSimple()	Whether a geometry is simple	<code>SELECT ST_IsSimple(g);</code>
ST_IsValid()	Whether a geometry is valid	<code>SELECT ST_IsValid(g);</code>
ST_LatFromGeoHash()	Return latitude from geohash value	<code>SELECT ST_LatFromGeoHash('ezs42');</code>
ST_Latitude()	Return latitude of Point	<code>SELECT ST_Latitude(g);</code>
ST_Length()	Return length of LineString	<code>SELECT ST_Length(g);</code>
ST_LineFromText(), ST_LineStringFromText()	Construct LineString from WKT	<code>SELECT ST_LineFromText('LINESTRING(0 0,1 1)');</code>
ST_LineFromWKB(), ST_LineStringFromWKB()	Construct LineString from WKB	<code>SELECT ST_LineFromWKB(g);</code>
ST_LineInterpolatePoint()	The point a given percentage along a LineString	<code>SELECT ST_LineInterpolatePoint(g,0.5);</code>
ST_LineInterpolatePoints()	The points a given percentage along a LineString	<code>SELECT ST_LineInterpolatePoints(g,0.5);</code>
ST_LongFromGeoHash()	Return longitude from geohash value	<code>SELECT ST_LongFromGeoHash('ezs42');</code>
ST_Longitude()	Return longitude of Point	<code>SELECT ST_Longitude(g);</code>
ST_MakeEnvelope()	Rectangle around two points	<code>SELECT ST_MakeEnvelope(0,0,1,1,4326);</code>
ST_MLineFromText(), ST_MultiLineStringFromText()	Construct MultiLineString from WKT	<code>SELECT ST_MLineFromText('MULTILINESTRING()');</code>
ST_MLineFromWKB(), ST_MultiLineStringFromWKB()	Construct MultiLineString from WKB	<code>SELECT ST_MLineFromWKB(g);</code>
ST_MPointFromText(), ST_MultiPointFromText()	Construct MultiPoint from WKT	<code>SELECT ST_MPointFromText('MULTIPOINT()');</code>
ST_MPointFromWKB(), ST_MultiPointFromWKB()	Construct MultiPoint from WKB	<code>SELECT ST_MPointFromWKB(g);</code>

Name	Description	Usage Example
ST_MPolyFromText(), ST_MultiPolygonFromText()	Construct MultiPolygon from WKT	<code>SELECT ST_MPolyFromText('MULTIPOLYGON()');</code>
ST_MPolyFromWKB(), ST_MultiPolygonFromWKB()	Construct MultiPolygon from WKB	<code>SELECT ST_MPolyFromWKB(g);</code>
ST_NumGeometries()	Return number of geometries in geometry collection	<code>SELECT ST_NumGeometries(g);</code>
ST_NumInteriorRing(), ST_NumInteriorRings()	Return number of interior rings in Polygon	<code>SELECT ST_NumInteriorRing(g);</code>
ST_NumPoints()	Return number of points in LineString	<code>SELECT ST_NumPoints(g);</code>
ST_Overlaps()	Whether one geometry overlaps another	<code>SELECT ST_Overlaps(a,b);</code>
ST_PointAtDistance()	The point a given distance along a LineString	<code>SELECT ST_PointAtDistance(g,10);</code>
ST_PointFromGeoHash()	Convert geohash value to POINT value	<code>SELECT ST_PointFromGeoHash('ezs42');</code>
ST_PointFromText()	Construct Point from WKT	<code>SELECT ST_PointFromText('POINT(0 0)');</code>
ST_PointFromWKB()	Construct Point from WKB	<code>SELECT ST_PointFromWKB(g);</code>
ST_PointN()	Return N-th point from LineString	<code>SELECT ST_PointN(g,1);</code>
ST_PolyFromText(), ST_PolygonFromText()	Construct Polygon from WKT	<code>SELECT ST_PolyFromText('POLYGON()');</code>
ST_PolyFromWKB(), ST_PolygonFromWKB()	Construct Polygon from WKB	<code>SELECT ST_PolyFromWKB(g);</code>
ST_Simplify()	Return simplified geometry	<code>SELECT ST_Simplify(g);</code>
ST_SRID()	Return spatial reference system ID for geometry	<code>SELECT ST_SRID(g);</code>
ST_StartPoint()	Start Point of LineString	<code>SELECT ST_StartPoint(g);</code>
ST_SwapXY()	Return argument with X/Y coordinates swapped	<code>SELECT ST_SwapXY(g);</code>
ST_SymDifference()	Return point set symmetric difference of two geometries	<code>SELECT ST_SymDifference(a,b);</code>
ST_Touches()	Whether one geometry touches another	<code>SELECT ST_Touches(a,b);</code>
ST_Transform()	Transform coordinates of geometry	<code>SELECT ST_Transform(g,4326);</code>
ST_Union()	Return point set union of two geometries	<code>SELECT ST_Union(a,b);</code>
ST_Validate()	Return validated geometry	<code>SELECT ST_Validate(g);</code>
ST_Within()	Whether one geometry is within another	<code>SELECT ST_Within(a,b);</code>
ST_X()	Return X coordinate of Point	<code>SELECT ST_X(g);</code>

Name	Description	Usage Example
ST_Y()	Return Y coordinate of Point	<code>SELECT ST_Y(g);</code>

Comparison Functions

Name	Description	Usage Example
COALESCE()	Return the first non-NULL argument	<code>SELECT COALESCE(col1, col2) FROM t;</code>
EXISTS()	Whether the result of a query contains any rows	<code>SELECT EXISTS(SELECT 1);</code>
NOT EXISTS()	Whether the result of a query contains no rows	<code>SELECT NOT EXISTS(SELECT 1);</code>
GREATEST()	Return the largest argument	<code>SELECT GREATEST(1,2,3);</code>
ISNULL()	Test whether the argument is NULL	<code>SELECT ISNULL(NULL);</code>
LEAST()	Return the smallest argument	<code>SELECT LEAST(1,2,3);</code>
NULLIF()	Return NULL if expr1 = expr2	<code>SELECT NULLIF(1,1);</code>

Control Flow Functions

Name	Description	Usage Example
CASE	Case operator	<code>SELECT CASE WHEN a > 0 THEN 'yes' ELSE 'no' END;</code>
IF()	If/else construct	<code>SELECT IF(a > 0, 'yes', 'no');</code>
IFNULL()	Null if/else construct	<code>SELECT IFNULL(a, 0);</code>

Information Functions

Name	Description	Usage Example
BENCHMARK()	Repeatedly execute an expression	<code>SELECT BENCHMARK(100000, MD5('a'));</code>
BIN_TO_UUID()	Convert binary UUID to string	<code>SELECT BIN_TO_UUID(UUID_TO_BIN(UUID()));</code>
BIT_COUNT()	Return the number of bits that are set	<code>SELECT BIT_COUNT(7);</code>
CONNECTION_ID()	Return the connection ID (thread ID) for the connection	<code>SELECT CONNECTION_ID();</code>
CURRENT_ROLE()	Return the current active roles	<code>SELECT CURRENT_ROLE();</code>
CURRENT_USER(), CURRENT_USER	The authenticated user name and host name	<code>SELECT CURRENT_USER();</code>
DATABASE()	Return the default (current) database name	<code>SELECT DATABASE();</code>
DEFAULT()	Return the default value for a table column	<code>SELECT DEFAULT(col);</code>
FOUND_ROWS()	For a SELECT with a LIMIT clause, the number of rows that would be returned were there no LIMIT clause	<code>SELECT FOUND_ROWS();</code>
ICU_VERSION()	ICU library version	<code>SELECT ICU_VERSION();</code>

Name	Description	Usage Example
LAST_INSERT_ID()	Value of the AUTOINCREMENT column for the last INSERT	<code>SELECT LAST_INSERT_ID();</code>
PS_CURRENT_THREAD_ID()	Performance Schema thread ID for current thread	<code>SELECT PS_CURRENT_THREAD_ID();</code>
PS_THREAD_ID()	Performance Schema thread ID for given thread	<code>SELECT PS_THREAD_ID();</code>
ROW_COUNT()	The number of rows updated	<code>SELECT ROW_COUNT();</code>
SCHEMA()	Synonym for DATABASE()	<code>SELECT SCHEMA();</code>
SESSION_USER()	Synonym for USER()	<code>SELECT SESSION_USER();</code>
SYSTEM_USER()	Synonym for USER()	<code>SELECT SYSTEM_USER();</code>
USER()	The user name and host name provided by the client	<code>SELECT USER();</code>
UUID()	Return a Universal Unique Identifier (UUID)	<code>SELECT UUID();</code>
UUID_SHORT()	Return an integer-valued universal identifier	<code>SELECT UUID_SHORT();</code>
UUID_TO_BIN()	Convert string UUID to binary	<code>SELECT UUID_TO_BIN(UUID());</code>
VERSION()	Return a string that indicates the MySQL server version	<code>SELECT VERSION();</code>
IS_UUID()	Whether argument is a valid UUID	<code>SELECT IS_UUID(UUID());</code>

Encryption and Compression

Name	Description	Usage Example
AES_DECRYPT()	Decrypt using AES	<code>SELECT AES_DECRYPT(col, 'key');</code>
AES_ENCRYPT()	Encrypt using AES	<code>SELECT AES_ENCRYPT('text', 'key');</code>
COMPRESS()	Return result as a binary string	<code>SELECT COMPRESS('text');</code>
RANDOM_BYTES()	Return a random byte vector	<code>SELECT RANDOM_BYTES(16);</code>
SHA2()	Calculate an SHA-2 checksum	<code>SELECT SHA2('text', 256);</code>
UNCOMPRESS()	Uncompress a string compressed	<code>SELECT UNCOMPRESS(col);</code>
UNCOMPRESSED_LENGTH()	Return the length of a string before compression	<code>SELECT UNCOMPRESSED_LENGTH(col);</code>
VALIDATE_PASSWORD_STRENGTH()	Determine strength of password	<code>SELECT VALIDATE_PASSWORD_STRENGTH('abc');</code>

Locking Functions

Name	Description	Usage Example
GET_LOCK()	Get a named lock	<code>SELECT GET_LOCK('my_lock', 10);</code>
IS_FREE_LOCK()	Whether the named lock is free	<code>SELECT IS_FREE_LOCK('my_lock');</code>
IS_USED_LOCK()	Whether the named lock is in use; return connection identifier if true	<code>SELECT IS_USED_LOCK('my_lock');</code>
RELEASE_ALL_LOCKS()	Release all current named locks	<code>SELECT RELEASE_ALL_LOCKS();</code>
RELEASE_LOCK()	Release the named lock	<code>SELECT RELEASE_LOCK('my_lock');</code>

Regular Expression Functions

Name	Description	Usage Example
REGEXP_INSTR()	Starting index of substring matching regular expression	<code>SELECT REGEXP_INSTR('abc123','[0-9]');</code>
REGEXP_LIKE()	Whether string matches regular expression	<code>SELECT REGEXP_LIKE('abc','^a');</code>
REGEXP_REPLACE()	Replace substrings matching regular expression	<code>SELECT REGEXP_REPLACE('abc123','[0-9]','');</code>
REGEXP_SUBSTR()	Return substring matching regular expression	<code>SELECT REGEXP_SUBSTR('abc123','[0-9]+');</code>

Replication Functions

Name	Description	Usage Example
asynchronous_connection_failover_add_managed()	Add group member source server configuration information to a replication channel source list	<code>SELECT asynchronous_connection_failover_add_managed('example');</code>
asynchronous_connection_failover_add_source()	Add source server configuration information server to a replication channel source list	<code>SELECT asynchronous_connection_failover_add_source('example');</code>
asynchronous_connection_failover_delete_managed()	Remove a managed group from a replication channel source list	<code>SELECT asynchronous_connection_failover_delete_managed('example');</code>
asynchronous_connection_failover_delete_source()	Remove a source server from a replication channel source list	<code>SELECT asynchronous_connection_failover_delete_source('example');</code>
asynchronous_connection_failover_reset()	Remove all settings relating to group replication asynchronous failover	<code>SELECT asynchronous_connection_failover_reset('example');</code>
group_replication_disable_member_action()	Disable member action for event specified	<code>SELECT group_replication_disable_member_action('example');</code>

Name	Description	Usage Example
group_replication_enable_member_action()	Enable member action for event specified	<pre>SELECT group_replication_enable_member_action('example');</pre>
group_replication_get_communication_protocol()	Get version of group replication communication protocol currently in use	<pre>SELECT group_replication_get_communication_protocol('example');</pre>
group_replication_get_write_concurrency()	Get maximum number of consensus instances currently set for group	<pre>SELECT group_replication_get_write_concurrency('example');</pre>
group_replication_reset_member_actions()	Reset all member actions to defaults and configuration version number to 1	<pre>SELECT group_replication_reset_member_actions('example');</pre>
group_replication_set_as_primary()	Make a specific group member the primary	<pre>SELECT group_replication_set_as_primary('example');</pre>
group_replication_set_communication_protocol()	Set version for group replication communication protocol to use	<pre>SELECT group_replication_set_communication_protocol('example');</pre>
group_replication_set_write_concurrency()	Set maximum number of consensus instances that can be executed in parallel	<pre>SELECT group_replication_set_write_concurrency('example');</pre>
group_replication_switch_to_multi_primary_mode()	Changes the mode of a group running in single-primary mode to multi-primary mode	<pre>SELECT group_replication_switch_to_multi_primary_mode('example');</pre>
group_replication_switch_to_single_primary_mode()	Changes the mode of a group running in multi-primary mode to single-primary mode	<pre>SELECT group_replication_switch_to_single_primary_mode('example');</pre>
MASTER_POS_WAIT()	Block until the replica has read and	<pre>SELECT MASTER_POS_WAIT('example');</pre>

Name	Description	Usage Example
	applied all updates up to the specified position	
SOURCE_POS_WAIT()	Block until the replica has read and applied all updates up to the specified position	<pre>SELECT SOURCE_POS_WAIT('example');</pre>
WAIT_FOR_EXECUTED_GTID_SET()	Wait until the given GTIDs have executed on the replica.	<pre>SELECT WAIT_FOR_EXECUTED_GTID_SET('example');</pre>

XML Functions

Name	Description	Usage Example
ExtractValue()	Extract a value from an XML string using XPath notation	<pre>SELECT ExtractValue('<a>1', '/a');</pre>
UpdateXML()	Return replaced XML fragment	<pre>SELECT UpdateXML('<a>1', '/a', '2');</pre>

Miscellaneous Functions

Name	Description	Usage Example
ANY_VALUE()	Suppress ONLY_FULL_GROUP_BY value rejection	<pre>SELECT ANY_VALUE(col) FROM t GROUP BY id;</pre>
BINARY	Cast a string to a binary string	<pre>SELECT BINARY('abc');</pre>
CAST()	Cast a value as a certain type	<pre>SELECT CAST('1' AS UNSIGNED);</pre>
CONVERT()	Cast a value as a certain type	<pre>SELECT CONVERT('1', UNSIGNED);</pre>
CONVERT_TZ()	Convert from one time zone to another	<pre>SELECT CONVERT_TZ(NOW(), 'UTC', 'SYSTEM');</pre>
ETAG()	Compute hash for each row, using one or more column or other values	<pre>SELECT ETAG('a');</pre>
FORMAT_BYTES()	Convert byte count to value with units	<pre>SELECT FORMAT_BYTES(1024);</pre>
FORMAT_PICO_TIME()	Convert time in picoseconds to value with units	<pre>SELECT FORMAT_PICO_TIME(1000);</pre>
FROM_BASE64()	Decode base64 encoded string and return result	<pre>SELECT FROM_BASE64('YQ==');</pre>
INET_ATON()	Return the numeric value of an IP address	<pre>SELECT INET_ATON('127.0.0.1');</pre>

Name	Description	Usage Example
INET_NTOA()	Return the IP address from a numeric value	<code>SELECT INET_NTOA(2130706433);</code>
MATCH()	Perform full-text search	<code>SELECT MATCH(col) AGAINST('term');</code>
NAME_CONST()	Cause the column to have the given name	<code>SELECT NAME_CONST('a',1);</code>
ROLES_GRAPHML()	Return a GraphML document representing memory role subgraphs	<code>SELECT ROLES_GRAPHML();</code>
SLEEP()	Sleep for a number of seconds	<code>SELECT SLEEP(1);</code>
STATEMENT_DIGEST()	Compute statement digest hash value	<code>SELECT STATEMENT_DIGEST('SELECT 1');</code>
STATEMENT_DIGEST_TEXT()	Compute normalized statement digest	<code>SELECT STATEMENT_DIGEST_TEXT('SELECT 1');</code>
STRING_TO_VECTOR()	Get the binary value of the VECTOR column represented by a conforming string	<code>SELECT STRING_TO_VECTOR('[1,2,3]');</code>
TO_BASE64()	Return the argument converted to a base-64 string	<code>SELECT TO_BASE64('a');</code>
VALUES()	Define the values to be used during an INSERT	<code>INSERT INTO t (c) VALUES (1);</code>
VECTOR_DIM()	Get the length of a vector (number of entries)	<code>SELECT VECTOR_DIM(vec);</code>
VECTOR_TO_STRING()	Get the string representation of a VECTOR column, given its value as a binary string	<code>SELECT VECTOR_TO_STRING(vec);</code>
CAN_ACCESS_COLUMN()	Internal use only	<code>SELECT CAN_ACCESS_COLUMN('example');</code>
CAN_ACCESS_DATABASE()	Internal use only	<code>SELECT CAN_ACCESS_DATABASE('example');</code>
CAN_ACCESS_TABLE()	Internal use only	<code>SELECT CAN_ACCESS_TABLE('example');</code>
CAN_ACCESS_USER()	Internal use only	<code>SELECT CAN_ACCESS_USER('example');</code>
CAN_ACCESS_VIEW()	Internal use only	<code>SELECT CAN_ACCESS_VIEW('example');</code>
GET_DD_COLUMN_PRIVILEGES()	Internal use only	<code>SELECT GET_DD_COLUMN_PRIVILEGES('example');</code>
GET_DD_CREATE_OPTIONS()	Internal use only	<code>SELECT GET_DD_CREATE_OPTIONS('example');</code>
GET_DD_INDEX_SUB_PART_LENGTH()	Internal use only	<code>SELECT GET_DD_INDEX_SUB_PART_LENGTH('example');</code>
INTERNAL_AUTO_INCREMENT()	Internal use only	<code>SELECT INTERNAL_AUTO_INCREMENT('example');</code>
INTERNAL_AVG_ROW_LENGTH()	Internal use only	<code>SELECT INTERNAL_AVG_ROW_LENGTH('example');</code>
INTERNAL_CHECK_TIME()	Internal use only	<code>SELECT INTERNAL_CHECK_TIME('example');</code>
INTERNAL_CHECKSUM()	Internal use only	<code>SELECT INTERNAL_CHECKSUM('example');</code>
INTERNAL_DATA_FREE()	Internal use only	<code>SELECT INTERNAL_DATA_FREE('example');</code>
INTERNAL_DATA_LENGTH()	Internal use only	<code>SELECT INTERNAL_DATA_LENGTH('example');</code>
INTERNAL_DD_CHAR_LENGTH()	Internal use only	<code>SELECT INTERNAL_DD_CHAR_LENGTH('example');</code>
INTERNAL_GET_COMMENT_OR_ERROR()	Internal use only	<code>SELECT INTERNAL_GET_COMMENT_OR_ERROR('example');</code>
INTERNAL_GET_ENABLED_ROLE_JSON()	Internal use only	<code>SELECT INTERNAL_GET_ENABLED_ROLE_JSON('example');</code>

Name	Description	Usage Example
INTERNAL_GET_HOSTNAME()	Internal use only	<code>SELECT INTERNAL_GET_HOSTNAME('example');</code>
INTERNAL_GET_USERNAME()	Internal use only	<code>SELECT INTERNAL_GET_USERNAME('example');</code>
INTERNAL_GET_VIEW_WARNING_OR_ERROR()	Internal use only	<code>SELECT INTERNAL_GET_VIEW_WARNING_OR_ERROR('example');</code>
INTERNAL_INDEX_COLUMN_CARDINALITY()	Internal use only	<code>SELECT INTERNAL_INDEX_COLUMN_CARDINALITY('example');</code>
INTERNAL_INDEX_LENGTH()	Internal use only	<code>SELECT INTERNAL_INDEX_LENGTH('example');</code>
INTERNAL_IS_ENABLED_ROLE()	Internal use only	<code>SELECT INTERNAL_IS_ENABLED_ROLE('example');</code>
INTERNAL_IS_MANDATORY_ROLE()	Internal use only	<code>SELECT INTERNAL_IS_MANDATORY_ROLE('example');</code>
INTERNAL_KEYS_DISABLED()	Internal use only	<code>SELECT INTERNAL_KEYS_DISABLED('example');</code>
INTERNAL_MAX_DATA_LENGTH()	Internal use only	<code>SELECT INTERNAL_MAX_DATA_LENGTH('example');</code>
INTERNAL_TABLE_ROWS()	Internal use only	<code>SELECT INTERNAL_TABLE_ROWS('example');</code>
INTERNAL_UPDATE_TIME()	Internal use only	<code>SELECT INTERNAL_UPDATE_TIME('example');</code>